

Model Development Phase Template

Date	20 June 2025
Team ID	SWTID1749707061
Project Title	Online Payments Fraud Detection using Machine Learning
Maximum Marks	5 Marks

Model Selection Report

In the model selection report for future deep learning and computer vision projects, various architectures, such as CNNs or RNNs, will be evaluated. Factors such as performance, complexity, and computational requirements will be considered to determine the most suitable model for the task at hand.

Model Selection Report:

Model	Description	Hyperparameters	Performance Metric (e.g . , Accuracy , F1 Score)
Random Forest	Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their outputs to make more accurate and stable predictions. It reduces overfitting by averaging results from different trees and works well with large datasets. In fraud detection, Random Forest is effective because it can handle complex data patterns and class imbalances.	-	Accuracy Score = 99.99%
Decision Tree	A Decision Tree is a model that splits data into branches based on feature values, leading to	-	Accuracy Score = 99.97%

	decisions at the leaves. It's simple, easy to interpret, and effective for classification tasks like fraud detection.		
Extra Tree	An Extra Tree (Extremely Randomized Tree) is an ensemble learning method similar to a Random Forest, but it introduces more randomness by selecting split points randomly rather than choosing the best split. This often makes it faster and can reduce overfitting, improving performance on high-dimensional datasets like those used in fraud detection.	-	Accuracy Score = 99.97%
Support Vector Machine	A Support Vector Machine (SVM) is a supervised learning algorithm used for classification and regression. It works by finding the best hyperplane that separates data points of different classes with the maximum margin. SVM is effective in high-dimensional spaces and is commonly used in fraud detection for its accuracy and ability to handle complex patterns.	-	Accuracy Score = 99.06%
XG Boost	XGBoost (Extreme Gradient Boosting) is a powerful and efficient machine learning algorithm based on gradient boosting. It builds an ensemble of decision trees sequentially, where each tree corrects the errors of the previous ones. Known for its speed and performance, XGBoost handles missing values well and is widely used in fraud detection tasks for its high accuracy and ability to manage large, complex datasets.	-	Accuracy Score = 99.97%